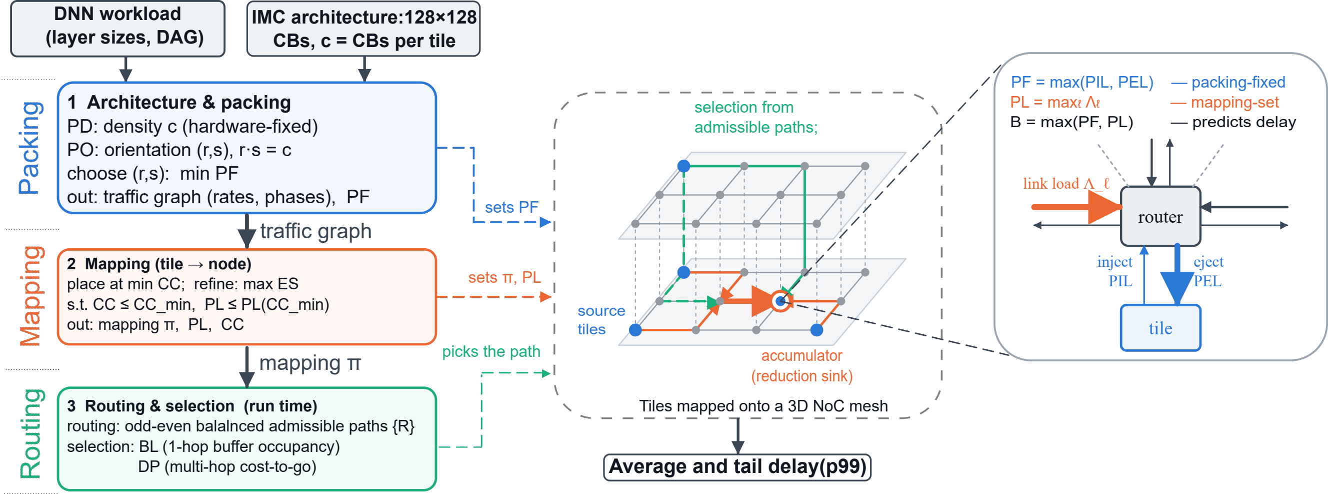

DNN workload (layer sizes, DAG)
IMC architecture:128×128 CBs, c = CBs per tile
Packing
1 Architecture & packing
PD: density c (hardware-fixed)
PO: orientation (r,s), r·s = c
choose (r,s): min PF
out: traffic graph (rates, phases), PF
sets PF
traffic graph
Mapping
2 Mapping (tile → node)
place at min CC; refine: max ES
s.t. CC ≤ CC_min, PL ≤ PL(CC_min)
out: mapping π, PL, CC
sets π, PL
mapping π
picks the path
Routing
3 Routing & selection (run time)
routing: odd-even balalnced admissible paths {R}
selection: BL (1-hop buffer occupancy)
DP (multi-hop cost-to-go)
selection from admissible paths;
source tiles
accumulator (reduction sink)
Tiles mapped onto a 3D NoC mesh
Average and tail delay(p99)
PF = max(PIL, PEL) — packing-fixed
PL = maxℓ Λℓ — mapping-set
B = max(PF, PL) — predicts delay
link load Λ_ℓ
router
inject PIL
eject PEL
tile